

PYTHON PROJECT BANK

STARTER LEVEL (15 pts)

1. Character Stats Generator

Project Prompt

Create a Python script that stores a character's name, class, and three stats. Print a "character card" that shows all the information clearly.

What You'll Make

A simple text-based character profile.

Core Skills You'll Practice

- Variables
- Strings and integers
- Printing text

What You Need to Know First

- Domain 1 Lesson 2: Strings & Integers
- Domain 3 Lesson 3: Print Formatted Text

Helpful Tools

Example starters:

- `name = "..."`
- `strength = ...`
- A quick list of your character's details before coding

Challenge Mode

Add a "power level" that adds all stats together.

2. Treasure Chest Calculator

Project Prompt

Write a program that calculates the total value of treasure based on gold, silver, and gems.

What You'll Make

A simple calculator that adds up treasure.

Core Skills You'll Practice

- Integers
- Arithmetic
- Printing results

What You Need to Know First

- Domain 1 Lesson 2: Numbers
- Domain 1 Lesson 7: Arithmetic Operators

Helpful Tools

- A small table to plan your values
- Example starter: `total = gold + silver + gems`

Challenge Mode

Add a multiplier for rare treasure.

3. Dialogue Printer

Project Prompt

Print a short conversation between two characters. Each line should print separately.

What You'll Make

A simple dialogue scene.

Core Skills You'll Practice

- Strings
- Escape characters
- Print formatting

What You Need to Know First

- Domain 1 Lesson 2: Strings
- Domain 3 Lesson 3: Print Formatted Text

Helpful Tools

- A short outline of the conversation
- A list of actions or emotions you want to include

Challenge Mode

Store character names in variables and reuse them.

4. Number Guessing Warm-Up (Single Guess)

Project Prompt

Choose a secret number in your code. Ask the user to guess it once. Tell them if they are right or wrong.

What You'll Make

A simple one-guess number game.

Core Skills You'll Practice

- Input
- Integers
- If/else

What You Need to Know First

- Domain 2 Lesson 1: Branching Statements
- Domain 1 Lesson 3: Data Type Conversion

Helpful Tools

A small diagram showing "guess → check → result"

Challenge Mode

Give a hint ("too high" or "too low").

5. Emoji Mood Reporter

Project Prompt

Ask the user how they feel and print an emoji that matches their mood.

What You'll Make

A mood-to-emoji converter.

Core Skills You'll Practice

- Input
- Strings
- If/elif/else

What You Need to Know First

- Domain 2 Lesson 1: Branching Statements
- Domain 1 Lesson 2: Strings

Helpful Tools

- A list of moods and matching emojis
- Example starter: `if mood == "happy":`

Challenge Mode

Add more moods and synonyms.

6. Basic Unit Converter

Project Prompt

Convert one unit to another (feet to meters, minutes to seconds, etc.).

What You'll Make

A simple conversion tool.

Core Skills You'll Practice

- Floats
- Arithmetic
- Input/output

What You Need to Know First

- Domain 1 Lesson 2: Numbers
- Domain 1 Lesson 7: Arithmetic Operators

Helpful Tools

- A list of formulas
- Sample inputs and outputs

Challenge Mode

Add multiple conversion options.

POWER-UP LEVEL (20 pts)

7. Adventure Path Selector

Project Prompt

Create a story where the player chooses between different options and sees different results based on their choices.

What You'll Make

A simple interactive story.

Core Skills You'll Practice

- If/elif/else
- User input
- Nested conditions

What You Need to Know First

- Domain 2 Lesson 1: Branching Statements
- Domain 3 Lesson 3: Console Input

Helpful Tools

- A simple map of your story choices
- A list of possible endings

Challenge Mode

Add hidden choices or extra endings.

8. Monster Battle Simulator

Project Prompt

Simulate a battle where the player attacks a monster and reduces its HP.

What You'll Make

A turn-based mini-battle.

Core Skills You'll Practice

- Variables
- Arithmetic
- Conditional logic

What You Need to Know First

- Domain 1 Lesson 7: Arithmetic Operators
- Domain 2 Lesson 1: Branching Statements
- Domain 2 Lesson 2: Loops (optional)

Helpful Tools

- A table for player and monster stats
- Example attack messages

Challenge Mode

Add a loop so the battle continues until someone reaches 0 HP.

9. Inventory Manager

Project Prompt Create a list of items and let the user add or remove items.	What You'll Make A basic inventory system.
Core Skills You'll Practice • Lists • Indexing • List operations	What You Need to Know First • Domain 1 Lesson 4: Indexing • Domain 1 Lesson 5: Lists
Helpful Tools • A simple inventory chart • A list of item ideas	Challenge Mode Add a limit to how many items the player can carry.

10. Loop-Based Workout Timer

Project Prompt Create a countdown or repeat an exercise name several times.	What You'll Make A simple loop-based timer.
Core Skills You'll Practice • For loops • While loops • Range	What You Need to Know First • Domain 2 Lesson 2: Iteration
Helpful Tools • A loop template • A sample countdown	Challenge Mode Add multiple exercises or rest periods.

11. Word Scrambler

Project Prompt Ask the user for a word, scramble the letters, and print the scrambled version.	What You'll Make A word-scrambling tool.
Core Skills You'll Practice • Indexing • Lists • Random	What You Need to Know First • Domain 1 Lesson 4: Indexing • Domain 6 Lesson 3: Random with Lists
Helpful Tools • A list of sample words • A step-by-step example of how scrambling works	Challenge Mode Let the user guess the original word.

12. Simple Shopping Cart

Project Prompt

Let the user add items to a cart and print the total.

What You'll Make

A basic cart system.

Core Skills You'll Practice

- Lists
- Loops
- Arithmetic

What You Need to Know First

- Domain 1 Lesson 5: Lists
- Domain 2 Lesson 2: Loops

Helpful Tools

- A list of sample items
- A price chart

Challenge Mode

Add item prices and calculate totals.

13. Password Strength Checker

Project Prompt

Ask the user for a password and check if it meets certain rules (length, numbers, symbols, etc.).

What You'll Make

A password checker.

Core Skills You'll Practice

- Strings
- Conditionals
- Searching text

What You Need to Know First

- Domain 2 Lesson 1: Branching
- Domain 1 Lesson 4: Indexing

Helpful Tools

- A list of password rules
- Examples of strong and weak passwords

Challenge Mode

Add a score or rating for the password.

LEGENDARY LEVEL (25 pts)

14. Dungeon Explorer

Project Prompt

Create a text-based dungeon where the player moves through rooms using a loop and makes choices that change what happens next.

What You'll Make

A multi-room adventure game.

Core Skills You'll Practice

- Loops
- Nested conditions
- Tracking player progress

What You Need to Know First

- Domain 2 Lesson 2: Iteration
- Domain 2 Lesson 3: Nested Loops
- Domain 3 Lesson 3: Console Input

Helpful Tools

- A simple map of your rooms
- A list of room descriptions

Challenge Mode

Add enemies, treasure, or a win/lose ending.

15. Save File System

Project Prompt

Save the player's name, level, and XP to a file — then load it when the program runs again.

What You'll Make

A basic save/load system.

Core Skills You'll Practice

- File input/output
- Data conversion
- Conditionals

What You Need to Know First

- Domain 3 Lesson 1: File Input and Output
- Domain 3 Lesson 2: Check Existence / Delete
- Domain 1 Lesson 3: Data Type Conversion

Helpful Tools

- A save file example
- A list of what you want to save

Challenge Mode

Add multiple save slots.

16. Guess the Creature (Advanced)

Project Prompt

The computer picks a creature from a list. The player guesses until they get it right.

What You'll Make

A guessing game with hints.

Core Skills You'll Practice

- Random
- Loops
- Lists

What You Need to Know First

- Domain 6 Lesson 3: Random
- Domain 2 Lesson 2: Loops
- Domain 1 Lesson 5: Lists

Helpful Tools

- A creature list
- Example hint messages

Challenge Mode

Add difficulty levels or a scoring system.

17. Text-Based Shop System

Project Prompt

Create a shop where players can buy items using gold and manage their inventory.

What You'll Make

A simple shop simulation.

Core Skills You'll Practice

- Lists
- Loops
- Conditionals
- Arithmetic

What You Need to Know First

- Domain 1 Lesson 5: Lists
- Domain 2 Lesson 2: Loops
- Domain 2 Lesson 1: Branching

Helpful Tools

- A price chart
- A list of shop items

Challenge Mode

Add random shop stock or limited quantities.

18. Mini-Encryption Tool

Project Prompt

Shift each letter in a message by a certain number to create a secret code.

What You'll Make

A simple encryption tool.

Core Skills You'll Practice

- Indexing
- Loops
- Strings

What You Need to Know First

- Domain 1 Lesson 4: Indexing
- Domain 2 Lesson 2: Loops

Helpful Tools

- An alphabet chart
- A sample encrypted message

Challenge Mode

Add a decryption mode.

19. High Score Tracker

Project Prompt

Store player scores in a file and print the top 5.

What You'll Make

A leaderboard that saves between runs.

Core Skills You'll Practice

- File I/O
- Sorting
- Lists

What You Need to Know First

- Domain 3 Lesson 1: File Input/Output
- Domain 1 Lesson 5: Lists

Helpful Tools

- A score table
- Example leaderboard

Challenge Mode

Add player names and timestamps.

20. Weather Forecaster (Randomized)

Project Prompt

Randomly generate weather for a fictional world and print a forecast.

What You'll Make

A random weather generator.

Core Skills You'll Practice

- Random
- Lists
- Conditionals

What You Need to Know First

- Domain 6 Lesson 3: Random
- Domain 2 Lesson 1: Branching

Helpful Tools

- A list of weather types
- Example forecast sentences

Challenge Mode

Add multi-day forecasts.

21. Function-Based Mini-Game

Project Prompt

Break a small game (battle, shop, or guessing game) into functions to keep your code organized.

What You'll Make

A modular program with clear sections.

Core Skills You'll Practice

- Functions
- Parameters
- Return values

What You Need to Know First

- Domain 4 Lesson 2: Function Definitions
- Domain 4 Lesson 2: Using return
- Domain 4 Lesson 2: Default Values

Helpful Tools

- A list of functions you plan to create
- A simple diagram showing how they connect

Challenge Mode

Add a main menu that calls different functions.